YASH SANTOSH PRASAD

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LinkedIn | Google Scholar

Education

Symbiosis Skills & Professional University – B. Tech Mechatronics Engineering (CGPA: 8.29) July 2026

Intellectual Property & Publications

• **Design Patent (Granted):** Integrated Communication Enabled Navigational Support Device (Design No.: 436426-001)

• **Utility Patent (Filed):** Mobility Aid with Advanced Navigation & Emergency Response (App No.: 202421073593)

• **Copyright (Registered):** Smart Mobility Aid With AI-Powered Navigation (Reg No.: LD-20250169851)

• **Design Patent (Granted):** Autonomous Modular Multi-Payload Aerial System (Design No.: 501462-001)

• **Utility Patent (Filed):** Autonomous Modular Multi-Payload Aerial System (App. No.: 202621059159)

• **Conference/Research Paper:** Autonomous Modular Multi-Payload Aerial System – Springer, CVR 2026, NIT Goa

• **Book Chapter:** Robotics Integration in Logistics through AS/RS – *Advances in Thermal Systems*

Projects

Aero Mind – AI Multipurpose Drone | ***Python, YOLOv8, Byte Track, Raspberry Pi 5, GPS, Lidar*** **Aug – Dec 2025**

- Engineered an edge-AI perception pipeline combining YOLOv8 and Byte Track on a Raspberry Pi 5, enabling real-time multi-object detection and tracking.
- Integrated autonomous GPS waypoint navigation and fail-safe return-to-home protocols, allowing fully hands-off field operations, and Lidar for Autonomous Air Obstacle Avoidance.
- Designed a quick-swap modular payload mechanism supporting three distinct operational modes: precision agricultural spraying, surveillance, and light parcel delivery

STM32MP1 Industrial Robot Communication System | ***Embedded Linux, Bash, USB-IP, TCP/IP*** ***June – Sep 2025***

- Architected a low-latency USB-over-IP communication bridge on the STM32MP1 MPU, allowing remote PC and mobile clients to control robotic actuators over a local network.
- Automated sequential network connection routines, engineering a finite-state machine (FSM) to initialize, authenticate, and pair network interfaces in a strict, predefined order.
- Built auto-recovery and failover logic, eliminating manual network setup and ensuring zero-touch reconnection during network drops in industrial environments.

Smart Assistive Device for Visually Impaired | Embedded System, Computer Vision, Sensors, IoT **Nov – Apr 2024**

- Designed and built an IoT-enabled smart mobility device integrating real-time obstacle detection via computer vision, automated emergency SOS alerts, and live GPS geolocation.
- Implemented multi-sensor fusion (ultrasonic/optical sensors) and haptic/audio feedback modules, achieving responsive hazard warnings with sub-second reaction latency.
- Authored IP & Research: Research directly resulted in 1 Patent, 1 Published Research Paper, and 1 Registered Copyright for novel assistive hardware and navigation software.

Internship Experience

Fiat Automobiles India Pvt. Ltd. **Pune, India**
Project Intern – Body Shop & Smart Manufacturing ***Jan 2026 – Jul 2026***

- Engineered a Safety LOTO (Lockout/Tagout) integration using RFID authentication and visual analytics/vision cameras around robotic welding cells to enforce zero-hazard operator safety protocols.
- Implemented Industry 4.0 IoT solutions and Poka-Yoke (error-proofing) mechanisms across the high-throughput automotive body shop line, mitigating assembly defects and downtime.

iTech Robotics & Automation Pvt. Ltd. **Pune, India**
R&D Intern ***Jun 2025 – Jul 2025***

- Optimized embedded firmware on the dual-core STM32MP1 MPU, streamlining memory allocation and thread communication.
- Assembled a modular drone platform engineered for heavy payloads up to 3 kg, Conducted a 3-day technical workshop on *Autonomous Drone Systems* at MIT-WPU College

EasyReach Solutions Pvt. Ltd. **Pune, India**
IoT Intern ***Jun 2024 – Jul 2024***

- Engineered robust device-to-cloud IoT pipelines bridging local microcontroller sensor nodes to cloud backends via MQTT and HTTP REST APIs with zero data packet loss.
- Built interactive real-time telemetry dashboards, visualizing streaming machine health and sensor metrics to enable predictive fault monitoring.

Maheshwari International **Pune, India**
Mechatronics Engineer Intern ***Jun 2023 – Jul 2023***

- Reconditioned and calibrated CNC & VMC industrial machines, performing precision axis alignment, mechanical overhaul.
- Formulated systematic preventive maintenance and diagnostic workflows, reducing recurring hardware breakdowns and improving overall machine uptime by 15%.

Skills

Languages & Firmware: Python, Embedded C/C++, Bash, Ladder Logic
Industrial Automation & Hardware: PLC (Siemens & AB), HMI, SCADA, Ethernet/IP, Industrial Sensors, LOTO & Poka-Yoke
Embedded Systems & IoT: STM32 (STM32MP1), Raspberry Pi, Arduino, MQTT, Fundamental HTTP REST, Sensor Fusion
Simulation & CAD Tools: KiCAD, ANSYS, OpenCV, YOLOv8, Matplotlib, Pandas, NumPy
CAD & Simulation (Familiar / Working Knowledge): MATLAB (Plotting & Basic Simulink Modeling), CATIA V5 (3D Part Modelling & Drafting)

Certifications

- **e-Yantra** (IIT Bombay / eLSI): Embedded Systems and Robotics — Scored 90/100 in final technical examination.
- **Paper Presenter** (CVR 2026, NIT Goa): Awarded Certificate of Presentation for oral delivery of peer-reviewed research at the 6th International Conference on Computer Vision and Robotics.
- **Workshop Lead & Resource Person** (SSPU): Drone Technology Workshop — Mentored participants on UAV hardware, flight controllers, and avionics.
- **Workshop Organizer** (SSPU): Robotics in a Day: From Concept to Creation — Led logistics and hands-on robotics track.